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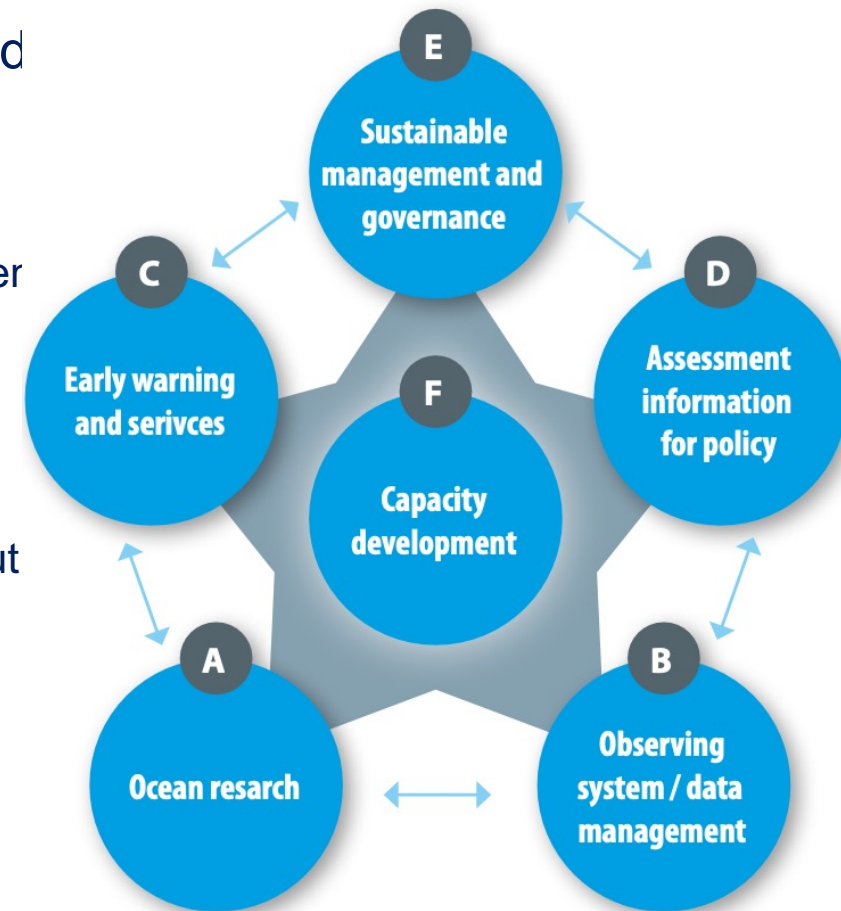
Intergovernmental
Oceanographic
Commission

**Delivering equitable and accessible
capacity development
through
coherence, coordination and
cooperation
in Capacity Development/Building
and Transfer of Marine Technology
(CBTMT)**

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United Nations Headquarters, New York
16-20 June 2025

IOC Capacity Development

- Capacity Development is a primary **Purpose** of the IOC, as included in its Statutes¹
- IOC CD is a stand alone core function of the IOC Medium-Term Strategy, 2022–2029, delivering its own initiatives e.g. Needs assessment surveys, Ocean Teacher Global Academy, Ocean CD-Hub, Ocean InfoHub, OBIS, ODIS
- IOC CD cross-cuts all other IOC functions:
 - All functions should adopt and deliver IOC CD Strategy as a key output
 - All Regional Offices should adopt and deliver IOC CD Strategy as a key output
- Collaborative efforts² and complementary activities by other UN organisations will be critical in ensuring cohesive and all-inclusive capacity development



¹ Article 2 – Purpose - 1. The purpose of the Commission is to promote international cooperation and to coordinate programmes in research, services and **capacity-building** [..]

² Article 2 – Purpose - 2. The Commission will **collaborate with international organizations** concerned with the work of the Commission, and **especially with those organizations of the United Nations** [..]

COHERENCE:

IOC Capacity Development Strategy

Output 1:

Human resources developed at individual and institutional levels

Output 2:

Access to technology, physical infrastructure, data and information established or improved

Output 3:

Global, regional and sub-regional mechanisms strengthened

Output 4:

Development of ocean research policies to support sustainable development objectives

Output 5:

Visibility, awareness and understanding on the roles and values of the ocean and ocean research in relation to human wellbeing and sustainable development increased

Output 6:

Sustained resource mobilization reinforced

Output	Activity	Action
1. Human resources developed at individual and institutional levels	1.1 Academic and higher education	1.1.1 Promote and assist with the strengthening and establishment of consortia of higher education and research institutions at the appropriate geographical scale
		1.1.2 Promote collaboration between UNESCO Chairs and IOC, and between IOC and other organisations dealing with ocean matters on human resources development
	1.2 Continuous professional development	1.2.1 Promote and assist with the organisation of training courses, workshops and 'summer schools', relevant to the IOC mandate, including training of trainers/technicians and executive career development for institutional managers/decision makers, in collaboration with other organisations
		1.2.2 Establish, or collaborate with other organisations to develop internship/fellowship and on-board training programmes
		1.2.3 Establish and collaborate with other organisations on visiting lecturer/scholar/researcher programmes and professional exchanges (peer to peer)
		1.2.4 Promote and assist with the establishment of regional training (and research) centres relevant to the IOC mandate
		1.2.5 Promote the development and sharing of training materials and tools

Output	Activity	Action
	1.3 Sharing of knowledge and expertise including through community building	1.3.1 Establish a travel grant "fund"
		1.3.2 Establish or collaborate with other organisations on a mentoring programme
		1.3.3 Promote and assist with the development and strengthening of IOC alumni networks, and professional networks including for youth leaders
		1.3.4 Promote funding, grant and scholarship programs to facilitate ocean research, technical development and scientific exchange through bringing visibility of opportunities via the Ocean CO-Hub
	1.4 Integration of ocean science in basic education	1.4.1 Promote the integration of ocean science in curricula of primary and secondary schools
	1.5 Improving gender, generational and geographic diversity	1.4.2 Promote careers in ocean research and ocean management
		1.5.1 Promote participation of women in ocean research
	2.1 Facilitating access to technology and infrastructure	1.5.2 Promote and support "young scientist" and "women in science" awards
		2.1.1 Establish and maintain a register of infrastructure to facilitate access, and promote transfer of marine technology
		2.1.2 Promote the development of regional collaboration on sustainable scientific infrastructure
	2.2 Facilitating equitable access to and sharing of ocean data and	2.1.3 Promote involvement of citizen science in ocean research
		2.2.1 Promote the development and wide use of regional and global data and information systems
		2.2.2 Promote the sharing of ocean data and

Output	Activity	Action
3. Global, regional and sub-regional mechanisms strengthened	3.1 Further strengthening and supporting secretariats of regional sub-commissions	3.1.1 Improve staffing of secretariat of regional sub-commissions
		3.1.2 Reinforce budgeting of regional sub-commissions
	3.2 Enhancing effective communication between regional sub-commission secretariats and global programmes as well as other communities of practice (incl. other organisations)	3.2.1 Establish an effective coordination and communication mechanism between the secretariats of the regional sub-commissions and the global programmes
		3.2.2 Establish an effective coordination and communication mechanism between the secretariats of the regional sub-commissions and the global programmes
	3.3 Identifying specific national and regional capacity development needs through regular needs assessment	3.3.1 Organise and conduct biennial capacity development needs survey
	3.4 Encouraging regional and sub-regional organisations to be leaders in, and amplifiers of capacity development	3.4.1 Reinforce engagement of regional and sub-regional organisations in consultation process and capacity development initiatives

Output	Activity	Action
4. Development of ocean research policies in support of sustainable development objectives promoted	4.1 Fostering the development of ocean research policies	4.1.1 Compile and compare information on existing ocean research policies, and disseminate to Member States for their use
		4.1.2 Assist and enable Member States with the development of ocean research policies, making use of the results of 4.1.1
5. Visibility, awareness and understanding on the roles and values of the ocean and ocean research in relation to human wellbeing and sustainable development increased	5.1 Fostering the development of ocean related public information and communication services	5.1.1 Encourage the development of public information (communication) departments in ocean research institutions
	5.2 Fostering the development of ocean literacy	5.2.1 Assist with the strengthening and development of ocean literacy programmes at national and regional levels
6. Sustained resource mobilization reinforced	6.1 Enhancing sustained support (in-kind and financial) to the IOC for its international coordination role	6.1.1 Foster partnerships to increase in-kind support opportunities
	6.2 Promoting sustained bilateral and multilateral support among Member States	6.2.1 Encourage resource mobilisation in particular from Member States, and other donors to support the outcomes of the IOC Capacity Development needs assessment

COHERENCE: IOC CD Strategy 2023-2030 Implementation Plan

Overarching Recommendations:

- 1) Leverage Existing Programs
- 2) Adapt a cohesive approach and improve implementation
- 3) Strengthen Regional Implementation
- 4) Supporting Global Coordination
- 5) Global Forum on Ocean Capacity Development
- 6) Strategic Partnerships
- 7) Strengthen Equity, Inclusion and Decolonialisation
- 8) ECOPs as Change-Makers
- 9) Enhanced Outreach & Communication
- 10) Prioritization & Risk Management
- 11) Monitor, Evaluate and Adapt



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IOC Capacity Development Strategy



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Международная
океанологическая
комиссия

Estrategia de desarrollo de capacidades
de la COI
Стратегия МОК в области развития
потенциала

2023-2030

Activity 1.2 Continuous professional development			
Action Action 1.2.1 Promote and assist with the organisation of training courses, workshops and "summer schools" relevant to the IOC mandate, including training of trainers/technicians and executive career development for institutional managers/decision makers, in collaboration with other organisations			
Objective - Enhance disciplinary, technical and managerial competencies among marine science professionals. - Build capacity through targeted training activities/programmes that are tailored to the needs of Member States and the IOC mandate. - Organize short-term, focused intensive training programmes (including summer schools) that focus on practical application and leadership development. - Develop competencies and implement learning frameworks related to Blue Skills, considering novel and emerging topics relevant to upskilling of ocean professionals and enhance institutions capability. - Enhance and leverage existing IOC projects and programs focused on capacity development			
Pathway - Identify specific technical, scientific, and management training needs by professionals via CD biennial survey and in conjunction with RSBs. - Implement a training request mechanism, e.g. official call or request form via OTGA platform - Design an integrated training program offered by IOC that includes courses, workshops, and summer schools aligned with identified needs. - Develop "training of trainers" (TOT) initiatives to ensure a cascading transfer of knowledge. - Further develop the CD-Hub as primary global marine science training portal for CD opportunities - Further develop the OceanTeacher Global Academy (Action 1.2.5) as a central online learning platform - Coordinate with consortia of higher education and potentially developed programs (Action 1.1.1). - Utilize the Capacity Development Facility's matchmaking service to replicate/scale up training initiatives and advance other capacity development activities prioritized by IOC			
Main Partners All IOC programs and initiatives, RSBs, International Oceanographic Data and Information Exchange (IODE), OceanTeacher Global Academy (OTGA), CDF, CD National Focal Points (Member States), ECOPs, UN and non-UN Organizations, Universities, NGOs		Resources Existing resources/New resources Existing and new partners	
		1.1.2 Promote collaboration between UNESCO Chairs and IOC, and between IOC and other organizations dealing with ocean matters on human resources development	- Support the Strengthening of the GOOS Africa coordination office at CURAT, Abidjan, Cote d'Ivoire
	1.2 Continuous professional development	1.2.1 Promote and assist with the organization of training courses, workshops and	1.Coastal Hazard Early Warning Systems o A coastal hazard modeling and forecasting certification program
			1. Blue Schools SIDS network 2. A New Blue Curriculum 3. Working to increase ocean knowledge among
			High Priority: - On Marine Spatial Planning Medium Priority:
			1. Deep-Sea 2. Coastal Hazard Early Warning Systems

COHERENCE:

IOC Criteria and Guidelines on the Transfer of Marine Technology (2005)



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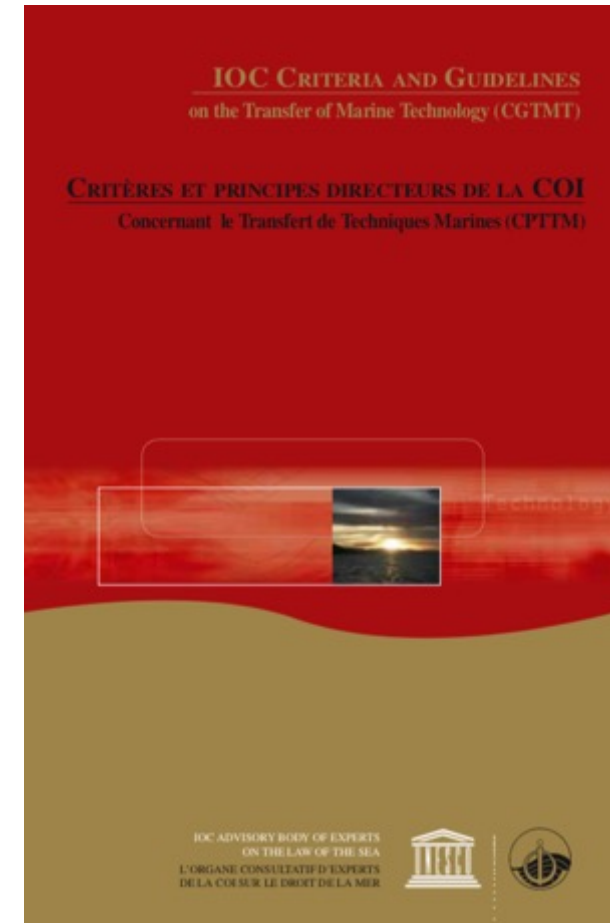
Definition of marine technology

Criteria:

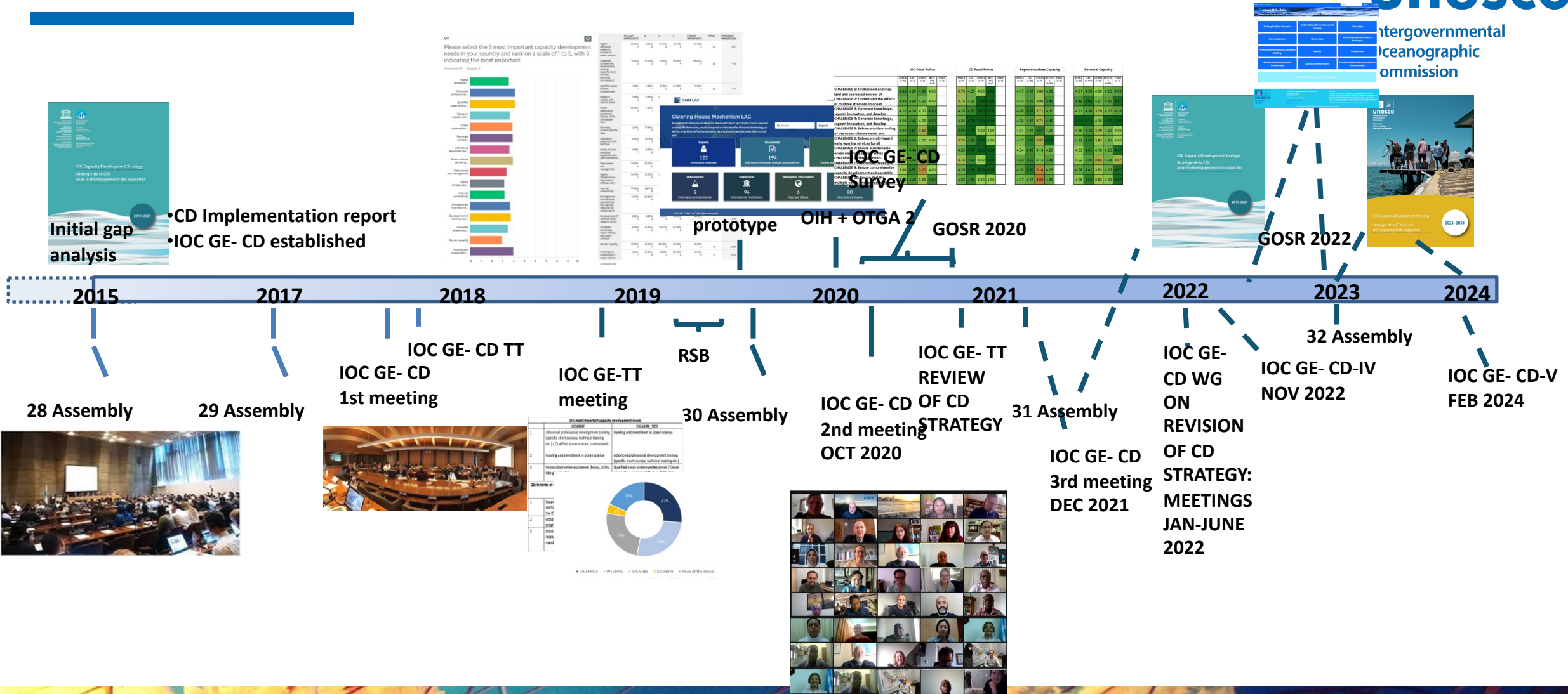
- 1) Developed schemes for TMT
- 2) Terms and Conditions
- 3) Interests and rights + environmentally sound approach
- 4) New cooperation schemes, partnerships, responsibility of IOC

Guidelines:

- 1) CHM
- 2) TMT as a part of national plans
- 3) Regional, sub-regional focal points
- 4) Promote TMT at IOC meetings
- 5) Seek financial contributions to TMT



Continuous expert mobilisation and engagement through the coordination of the IOC Group of Experts on Capacity Development



Clearing House Mechanism for Transfer of Marine Technology (CHM/TMT)



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Ocean Information Hub Global Search Portal

Provides improved access to ocean data, information, and knowledge products through a decentralized, interoperable system. Powered by ODIS.

oceaninfohub.org

130K content items

27K
marine
experts

13K
institutions

3,6K
projects

1,5K
training
assets

42K
documents

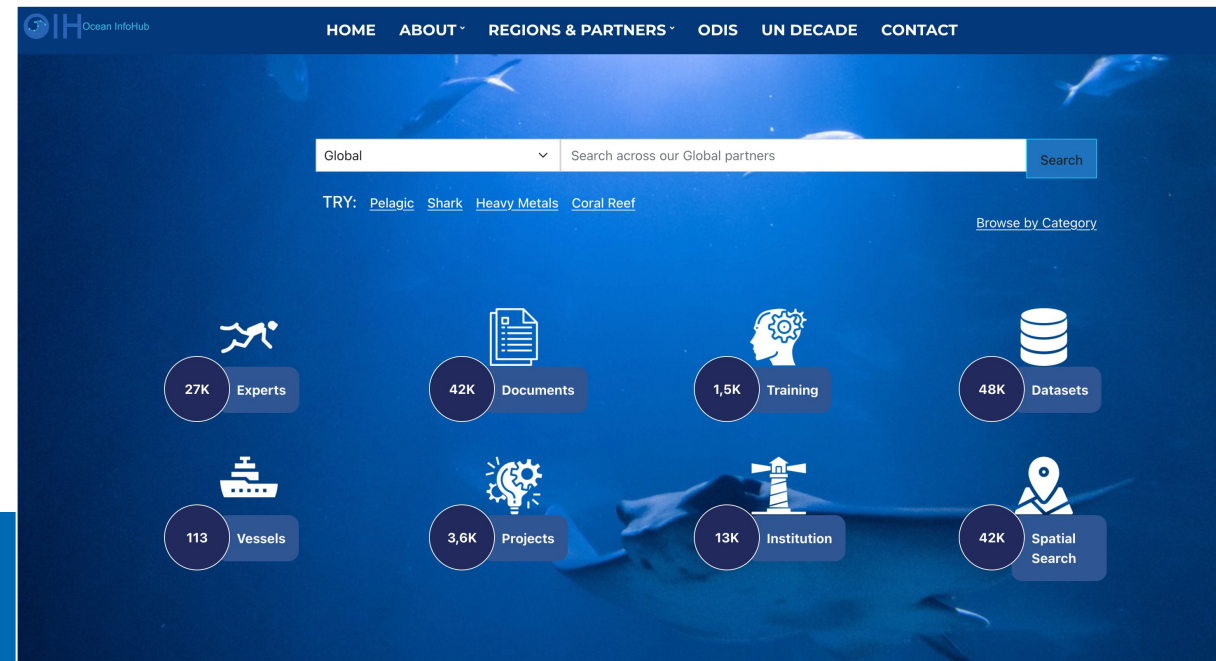
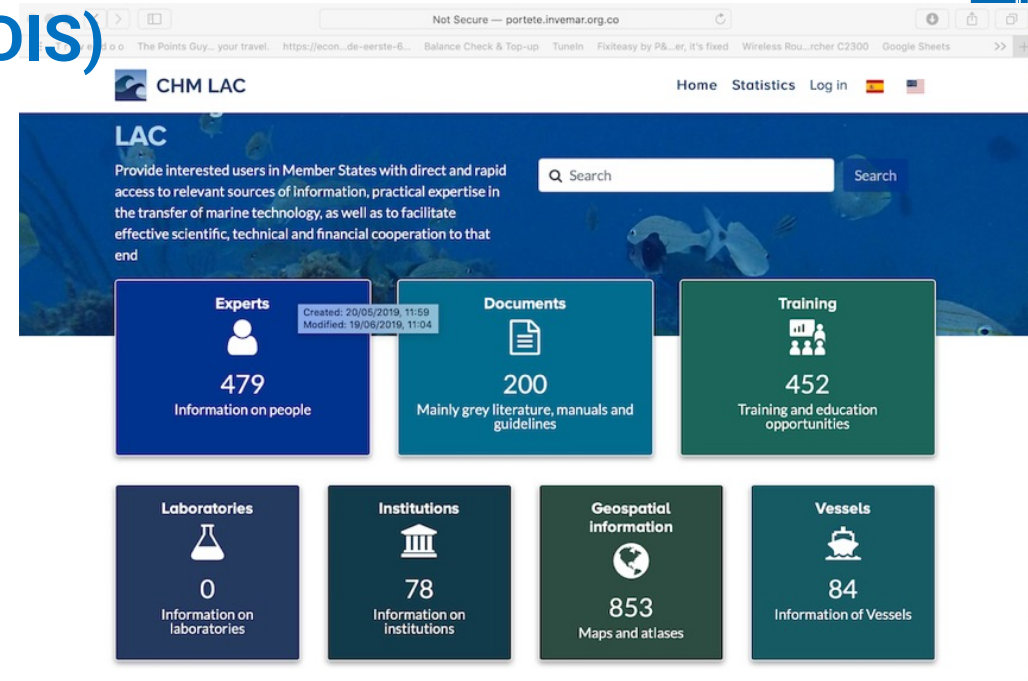
113
vessels

48K
datasets

Supports BBNJ Pillar 4: Capacity Building & Technology Transfer / OIH enables countries, especially developing states, LDCs, and SIDS, to connect with global information systems without needing to build their own from scratch.

Ocean Data and Information System (ODIS)

- A global search portal to discover the wealth of ocean data and information (Google for the Ocean).
- The central ODIS portal connects to regional/sectoral data portals (e.g, 3 regions: LAC, Africa and P-SIDS).
- A small project supported by NORAD on biodiversity Data Hub for the High Seas based on ODIS technology is ongoing. The hub can act as a demonstrator for several elements of the BBNJ CHM.



ODIS in the regions



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- **LAC:** INVEMAR has developed a regional node, which links ODIS to datasets from 12 countries in the LAC region as well as regional partners that include CLME+ Training and Capacity Development Portal, The Sargassum Information Hub and the Caribbean Marine Atlas.
- **PSIDS:** The Pacific has linked its two existing regional portals to OIH/ODIS which facilitates greater access to global marine data of Pacific relevance, that can be packaged and made available to SIDS in the Pacific.
- In **Africa**, The OceanInfoHub Project worked closely with IOC Africa and African network partners in the co-design process. In addition to the ODIS network itself, four new databases were established in cooperation with OIH/ODIS:

ODIS in the regions

Africa Marine Training Database (278 programmes/courses, 18 countries)

<https://africa.marinetraining.org/home-0>

IOC Africa Projects (177) <https://ioc-africa.org/dbs/advancedSearch.php?projects=all>

IOC Africa vessels (32) <https://ioc-africa.org/dbs/advancedSearch.php?vessels=all>

IOC Africa expeditions (117) https://iocafrica.org/dbs/advancedSearch.php?previous_upcoming=previous

•Africa node contributors: over 100 institutional sources contributing records via global nodes and regional nodes.

•**Global nodes** contributing data or information related to Africa include: OceanExpert, Aquadocs, OBIS, OBPS and others.

•**African nodes active** include Marine Spatial Atlas for the Western Indian Ocean of Coastal Oceans Research and Development in the Indian Ocean (CORDIO-MASPAWIO <http://maspawio.net/>), Benguela Current Convention (BCC), AfricaMarine Training <https://africa.marinetraining.org/home-0>, IOCAfrica databases mentioned above, and the African coastal and Marine Atlas catalogue (<https://acma.africanmarineatlas.org/#/>), Canary current LME Data Viewer; Marine Information Management system (South Africa); and the Western Indian Ocean Symphony for MSP.

••**NODCs/ ADUs** being supported to develop fundamental infrastructure (metadata catalogues) Tunisia, Namibia.

Capacity building and transfer of marine technology (CBTMT) services including data and information

1. Online data sharing platforms:

- International Oceanographic Data and Information Exchange (IODE)
- Ocean Data and Information System (ODIS)
- Ocean Biodiversity Information System (OBIS)

2. Monitoring and observation networks:

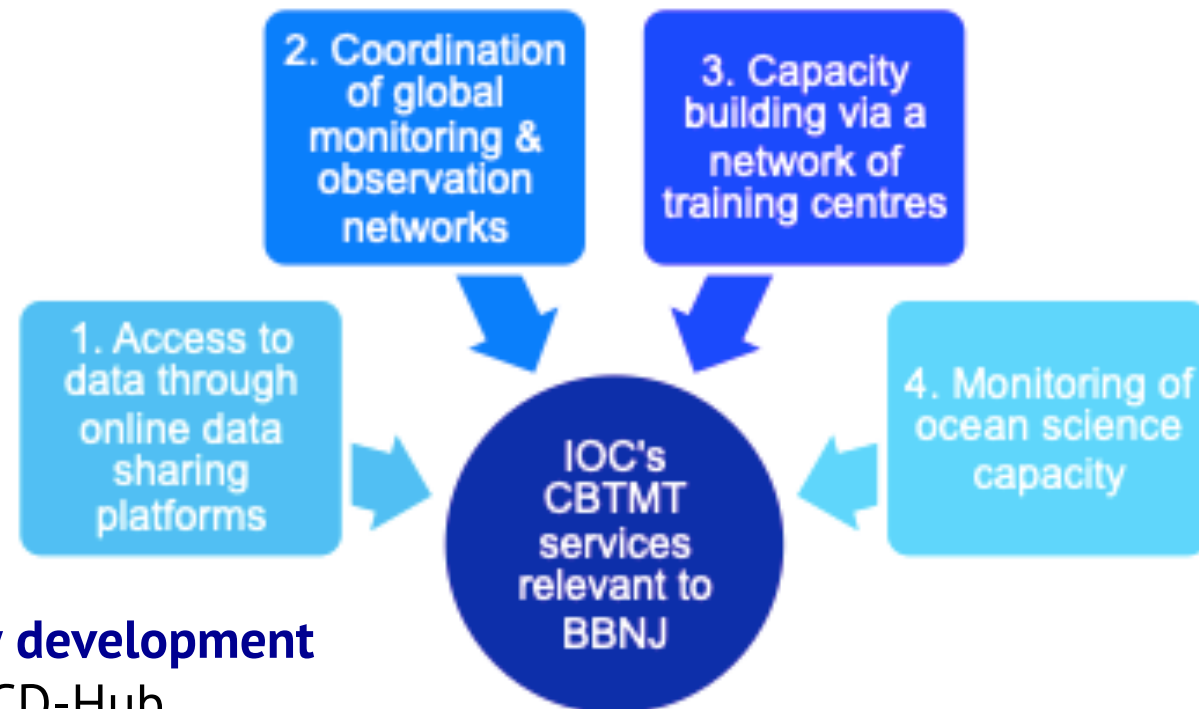
- Global Ocean Observing System (GOOS)
- Harmful Algal Information System (HAIS)
- Global Ocean Acidification Observing Network (GOA-ON) and ocean acidification data portal

3. Capacity development

- Ocean CD-Hub
- OceanTeacher Global Academy (OTGA)
- IOC Regional Network of Training and Research Centres

4. Monitoring ocean science capacity

- Global Ocean Science Report (GOSR)

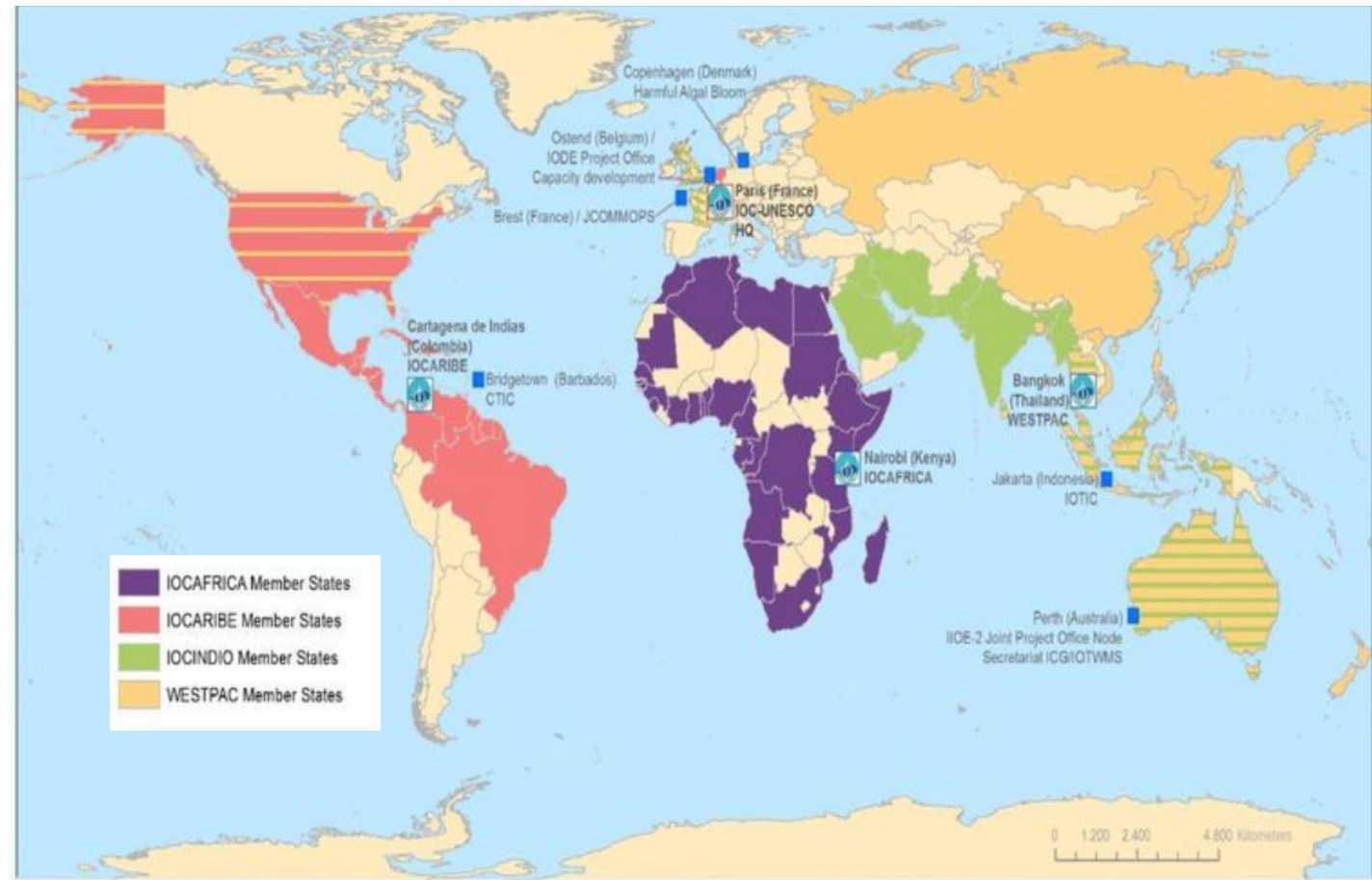


COORDINATION:

Regional coordination through regional sub-commissions

Strengthen Implementation within IOC

- IOC CD: **cross-cuts all other IOC functions**: all functions should adopt and deliver IOC CD Strategy and IP as a key output
- IOC CD: **Strategy and IP supportive of RSCs** (Output 3) and call for staffing and funding.
- RSCs are key in the successful delivery and implementation of a **cohesive and inclusive capacity development** in the regions
- RSCs should adopt and deliver IOC CD Strategy and IP as a key output
- **Regions are all different** and have varying CD needs and societal priorities
- Support/Improve **regional needs assessments**
- Develop **regional workplans and expert team** (Working Groups, Focal Points..), that responds to the IOC capacity development strategy and IP



COORDINATION: OceanTeacher Global Academy



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**Ocean Teacher Global
Academy**

IOC's global network of Regional and Specialised Training Centres, and affiliated partners empowers the global community through knowledge sharing and coordinated training.

oceanteacher.org

17
training
centres

100+
training
partners

15K
users
worldwide

187
countries
involved

300+
learning
sets

4
languages
available

Supports BBNJ Pillar 4: Capacity Building & Technology Transfer / Provides capacity to all nations to participate fully and benefit from the agreement, with direct contribution to Article 44 of BBNJ.



Our Alumni

Researchers
 Data Managers
 Natural Resource Managers
 Practitioners in operations and support
 Decision and policymakers
 Institutional managers
 Technicians, advisors and consultants
 Extension services
 Teachers and educators
ECOPs, graduates and postgraduates

Partners

International Organizations, programs, projects
 UN agencies



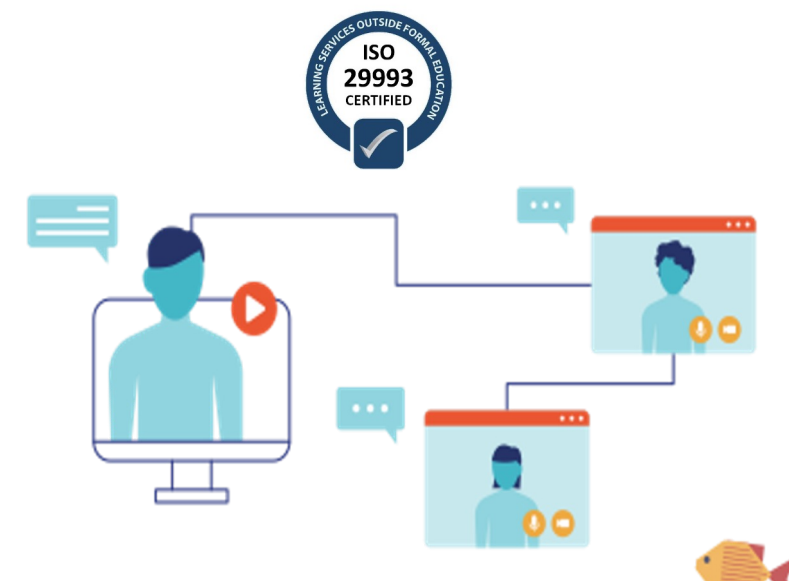
 Scientific Knowledge and Research Course topics include Research data management, OBIS, HAB	 Sustainable Use of Marine Resources Course topics include Marine Biodiversity Data Management	 Marine Spatial Planning Course topics include Marine spatial planning, GIS applications for ICZM	 Marine and Coastal Ecosystems Course topics include Marine GIS applications, Coastal mapping & monitoring
 Disaster Risk Reduction Course topics include Tsunami awareness, Storm surge forecasting	 Implement and Enforce International Sea Law Course topics include Marine Scientific Research under the UNCLOS	 Ocean Acidification Course topics include Impacts of ocean acidification	 Marine Pollution Course topics include Prevention and reduction of marine pollution





Learning Management System, services and products

- Planning, development, and delivery of training
- Coordination and management of courses and alumni.
- Shared e-Learning Platform
- Creation of learning resources and activities
- Mechanisms for connecting experts, facilitators, and centres
- Guidance on learning services procedures and standards
- Training and mentoring exchange and internships



ISO-certified learning management system

Automated reporting (KPIs)



BLUE SKILLS

Envisioning training programs to cover the key competencies expected of highly qualified ocean professionals.

Vision 2025-2030

CREATIVE EXPERIENCES

Preparing training infrastructure and learning resources to facilitate immersive and realistic practice.



SHARED SOLUTIONS

Promoting opportunities to develop and implement sustainable ocean solutions based on collaborative experiences, support frameworks, and cross-sectoral partnerships.



CORE SERVICES

Strengthen the capacity of the Regional and Specialized Training Centres to provide coordinated support and sustain high-quality impactful training programs.



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COORDINATION: Ocean Expert



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OceanExpert

Main repository for managing information on IOC events and their experts, institutes, and documents. Supports collaboration and open information between experts and networks.

oceanexpert.org

33K
active
experts

9,2K
active
institute
s

3,7K
events

35K+
accessible
documents

Supports BBNJ Pillar 4: Capacity Building & Technology Transfer / Facilitates networking, collaboration, and knowledge exchange among stakeholders worldwide. Contributes to building the human and institutional capacities necessary for effective implementation of the BBNJ Agreement.

COORDINATION: Ocean Capacity Development Hub (CD-Hub)



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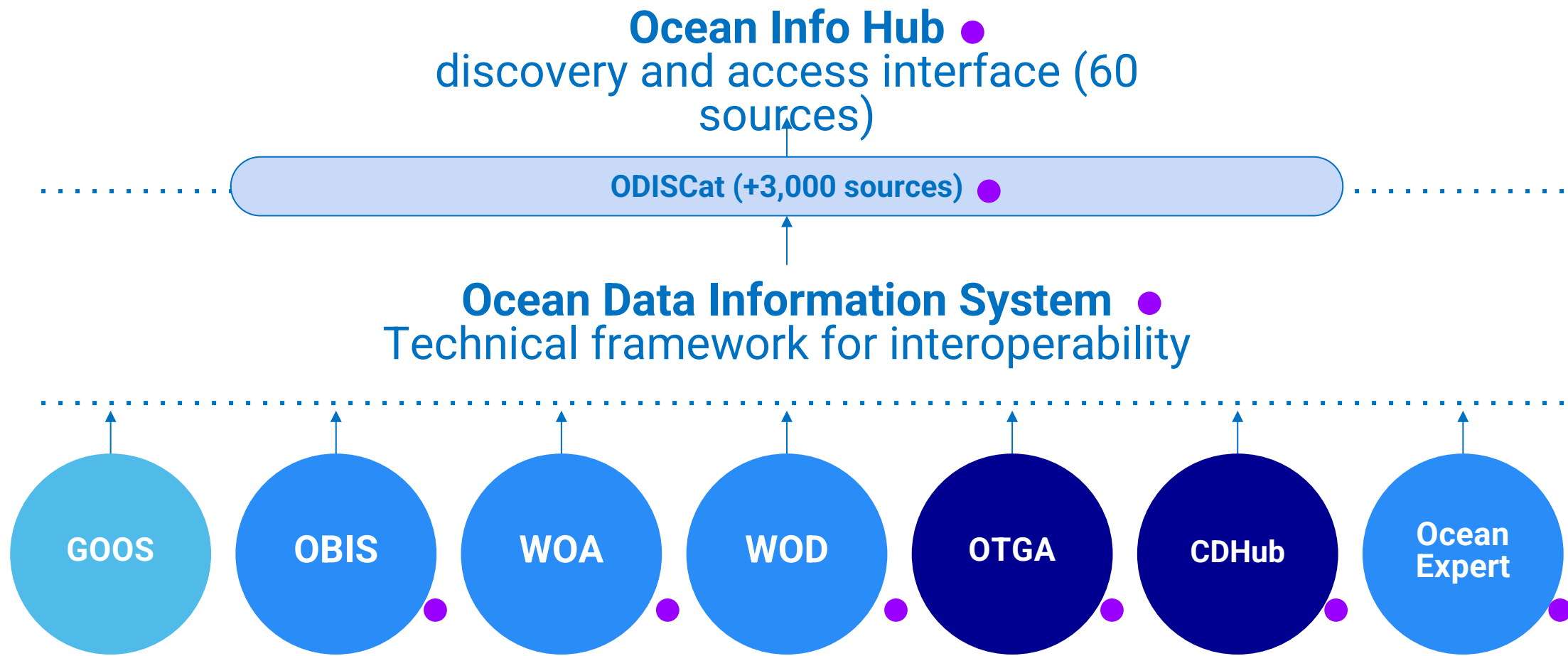
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 **Ocean CD-Hub**
Gateway to Ocean related Capacity Development opportunities around the world

Training & Higher Education	Onboard Expeditions, Research & Training	Fellowships
Internships/Jobs	Mentorships	Visiting Lecturers/Scholars & Exchanges Reciprocal or collaborative relations between senior and junior professional to foster growth, learning and career development
Professional Networks & Community Building	Awards	Travel Grants
Research Funding, Grants & Scholarships	Access to Infrastructure	Ocean Literacy, Public Information & Communication

COOPERATION: Overview of the operational IOC-UNESCO Clearing-House Mechanism



Part of IODE



Global networks coordination



Data access



Capacity development

Project: BlueCloud 2026



Blue-Cloud2026



EU-HORIZON



6 Courses on Virtual Research Environment

- Empowering Ocean Science through Virtual Research Environments
- VLab 1 - Mastering Coastal Ocean Observations with the Blue Cloud Virtual Labs
- VLab 2 - How to use a Virtual Laboratory on Coastal Ocean Currents from Observation
- VLab 3 - How to use a Virtual Laboratory on Carbon Plankton Dynamics
- VLab 4 - How to use a Virtual Laboratory on Marine Environmental Indicators
- VLab 5 - Navigating the Global Fisheries Atlas Virtual Lab: Tools and Techniques

Project: ADAPT



2 Courses in the Caribbean and 2 Online Courses

- ADAPT Curso de formación sobre mejores prácticas oceánicas
- ADAPT Training Course about ocean best practices
- Collaborative development of guidelines on ocean observation and best practices, using the Caribbean as a pilot region
- IOCARIBE leading



Proposals: International Coordination Action



International Coordination Action - OceanConnectXL

- Support ECOPs in developing Blue skills and competencies needed for international ocean research and policy-research in ocean management by creating training opportunities in Flanders;
- Develop mechanisms for international research collaboration through community building among ECOPs and EsCOP.

Proposals: M.A.R.I.S. - Marine and Aquatic Research and Investigation Schools



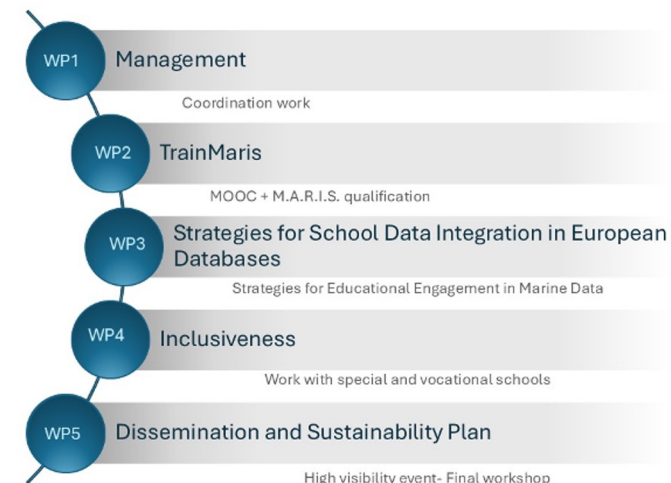
Erasmus+

Promoting Education for Sustainable Development and Environmental Science in European Classrooms

- Support ECOPs in developing Blue skills and competencies needed for international ocean research and policy-research in ocean management by creating training opportunities in Flanders;
- Develop mechanisms for international research collaboration through community building among ECOPs and EsCOP.

Marine and
Aquatic
Research and
Investigation
Schools

MARIS



Proposals: Training courses in OceanLiteracy



UNIVERSIDADE
TÉCNICA DO
ATLÂNTICO



CAMPUS
DO MAR



SGP

The GEF
Small Grants
Programme



Adaptation of training courses and materials in support of ocean educators

- Training guidance and roadmap
- Learning kits
- Citizen science projects



Project: OceanTraining Internships



International Traineeship to enhance global human capacity in the operationalization of information systems and delivery of specialized training

- 2 Calls for Host institutions and Mentors
- 2 Cohorts of trainees



Ocean Biodiversity Information System



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Ocean Biodiversity Information System

Provides globally accessible marine biodiversity data (FAIR), supports access & benefit-sharing.

obis.org

136M
species
observations

195K
marine
species

36
Nodes
worldwide

27M
DNA
sequences

6K
scientists &
data managers
involved

99
countries
engaged

Supports BBNJ Pillar 1: Marine genetic resources, including the fair and equitable sharing of benefits / Data repository supporting data transparency and equitable access to MGR information; helps fulfill obligations under access and benefit-sharing provisions (could provide SBIs, support monitoring and transparency). Support in data management plans.

Ocean Data and Information System



Ocean Data and Information System

Provides the interconnected technical backbone that makes marine data FAIR, and serves as the technical foundation of the Ocean InfoHub (OIH) interface.

odis.org

ODIS Catalogue of Sources (ODISCat)

The “Yellow Pages” of marine data: searchable directory of websites and systems that provide ocean-related data and information.

Supports BBNJ Pillar 4: Capacity Building & Technology Transfer / The BBNJ agreement could leverage ODIS technology to build a “BBNJ Knowledge System”: an open, interoperable platform connecting data, tools, and training resources to ensure equitable access to marine knowledge and technology, especially for developing countries.

Leveraging the UN Ocean Decade for CBTMT



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BARCELONA CONFERENCE: Recommended Ocean Decade priorities to 2030 including those relevant to BBNJ implementation:

- Science and knowledge
- ILK and cultural values
- Infrastructure for observations and modeling
- Data sharing and management
- Capacity development

UNOC3 SIDE EVENT ON IOC CD-DECADE CDF COLLABORATION:

- Matchmaking Platform of the Capacity Development Facility demonstrated



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THANK YOU